TS76 Coolant Temperature Sensor – Observed Calibration Report

This report summarizes the *observed* resistance–temperature relationship of the TS76 coolant sensor, derived directly from your CoolTerm log. Values are unaltered from the original capture; the Steinhart–Hart fit is included only for comparison.

Steinhart–Hart Coefficients (reference fit)

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A = 1.3939762853e-03
B = 2.5365362353e-04
C = -3.3944983556e-08
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Display Formula (Fahrenheit)

$$T_{°F} = (1 / (A + B \cdot \ln(R) + C \cdot (\ln(R))^3) - 273.15) \times 9/5 + 32$$

Inline copy version: T°F = (1 / (A + B·ln(R) + C·(ln(R))³) – 273.15) × 9/5 + 32

Display Formula (Celsius)

$$T_{°C} = 1 / (A + B \cdot \ln(R) + C \cdot (\ln(R))^3) - 273.15$$

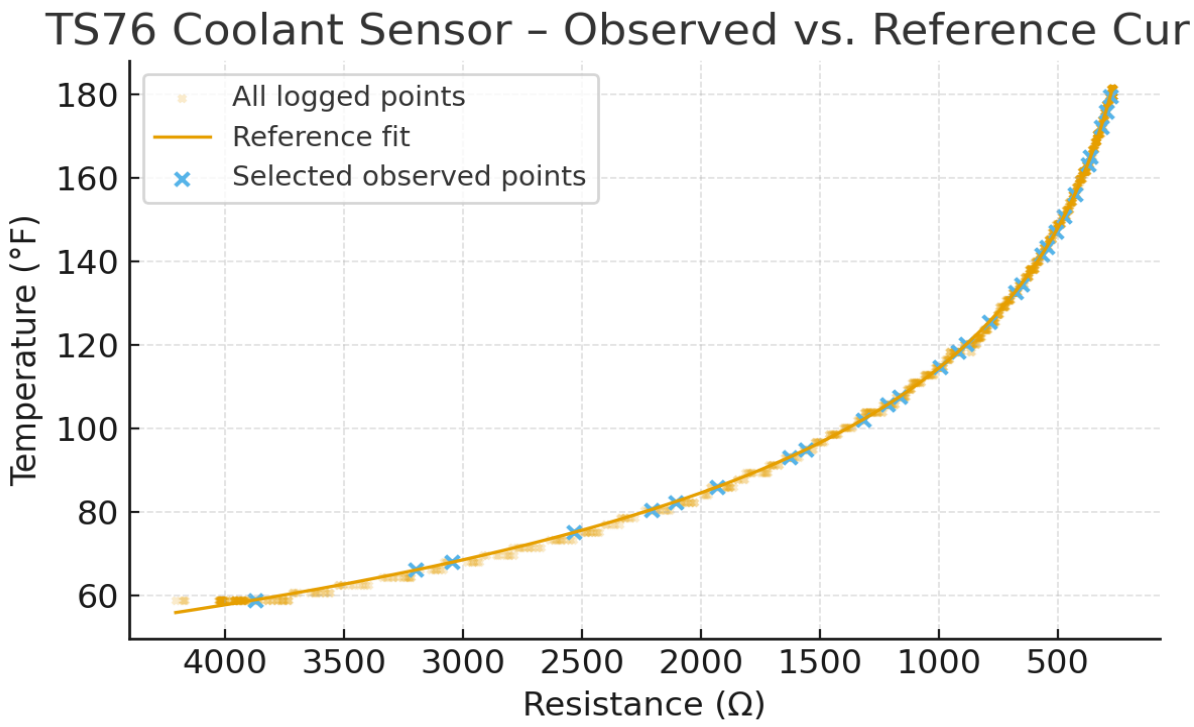
Inline copy version: T°C = 1 / (A + B·ln(R) + C·(ln(R))³) – 273.15

Representative Observed Data (from log)

Temp (°C)	Temp (°F)	Resistance (Ω) (observed)
15.0	59.0	3872
19.0	66.2	3198
20.0	68.0	3047
24.0	75.2	2534
27.0	80.6	2206
28.0	82.4	2104
30.0	86.0	1932
34.0	93.2	1627
35.0	95.0	1557
39.0	102.2	1317
41.0	105.8	1215
42.0	107.6	1165

46.0	114.8	994
48.0	118.4	918
49.0	120.2	884
52.0	125.6	786
56.0	132.8	678
57.0	134.6	652
61.0	141.8	565
62.0	143.6	545
64.0	147.2	508
66.0	150.8	474
69.0	156.2	427
73.0	163.4	373
74.0	165.2	361
78.0	172.4	316
80.0	176.0	297
82.0	179.6	279

Observed vs. Reference Curve



Notes

• Data source: direct CoolTerm capture from live PCM PID 05 and analog divider readings.
• Table values are observed (not computed).
• The Steinhart–Hart equation may be used for interpolation between recorded points.
• All resistance values are in ohms; temperature in degrees Celsius and Fahrenheit.